ASCII String Pattern Command Manual

Project	Content	Notes
Version	V1.0	
Release Date	August 20, 2018	
Scope of Application	Control the motion controller by sending string commands via the serial port. Suitable for scenarios not using binary protocols	
Related Reference	Secondary Development Manual	

Revision History

Version	Date	Content	Notes
V1.0	August 20, 2018	Completed the first version of the ASCII string instruction set specification, supporting Minimum compatible firmware version: V2.16.0.646	

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1 ASCII String Mode Command Overview

The controller/control card defaults to a binary communication format, encapsulating communication commands and providing corresponding dynamic link libraries. Generally, binary mode is recommended for development due to its advantages of faster communication speeds and broader hardware interface support.
ASCII string mode is suitable for scenarios where binary format is not used, offering the advantages of being easy to understand and integrate. The limitation is **that it must use a serial port and cannot use a USB port**.

1.1 Serial Port Format

The communication serial port format is **115200,N,8,1**

Baud rate: 115200
No parity
8 data bits
1 stop bit

1.2 Communication Process

Communication operates in **an acknowledgment-based half-duplex** mode, with each complete cycle consisting of one query and one response. The controller/control card functions as the controlled entity, while the user-end PC, industrial control board, touchscreen, etc., serve as the master controller.

For each valid command sent by the master, the controller/control card responds once. After evaluating the response content, the master proceeds with subsequent queries and responses as required by the application.
Note: If the command code is invalid, **the controller/control** card **will not respond** (the binary command parsing engine operates concurrently, and under certain conditions may conflict, resulting in no response). In such cases, the master side must implement **a timeout mechanism**, e.g., no response within 50ms indicates an invalid command code or other error.
For example:
Master: P_Abs 0 10000
Controlled Party: OK
The master issues a command to position the first axis to 10000 pulse units in absolute mode. The controller/control card responds and begins execution, accelerating at the set speed. The "OK" response indicates execution has started. Different

commands with different parameters result in varying completion times.

1.3 String Command Format

Command String Parameter 1 Parameter 2

Parameter 3…Enter/Line Break

XXXXX<space>n<space>n<space><\r\n>

1) All characters must be in half-width English mode. All subsequent characters must be English letters. Commands are case-insensitive. Parameters must be integer values supporting +and- signs but not decimal points. Units are pulse-based; physical quantities require proportional conversion by the master controller.

For example, for a linear stage with 1mm pitch and a 32-microstep stepper driver, 6400 pulses (200*32) correspond to 1mm. To move 2mm, set 6400*2=12800 pulses. Consult relevant personnel for specific controller parameters.

2) Strings are separated by spaces, where the ASCII code for space is 32.

3) Commands terminate with a carriage return followed by a line feed. The ASCII code for carriage return is 13, and for line feed is 10. This corresponds to the This ending is required for proper operation.

Response content must also end with a carriage return and line feed.

4) Subsequent detailed command descriptions will not explicitly mention spaces or CR/LF; compliance with this format is assumed.

5) For control concepts related to commands, refer to the terminology explanations in the secondary development manual.

6) Response Content List

Response	Description
OK	Command received correctly and execution initiated. Execution time and related parameters depend on the command and associated parameters. and related parameters
E1	Insufficient number of parameters
E2	Illegal characters exist in the parameters, or they do not meet the requirements of this instruction.
Numeric string	Returned read value, e.g., 1,345678, -3223, etc.

2 Command List and Routines

2.1 Handshake

2.1.1 CHECK

Item	Description	Remarks
Function	Detect whether the communication link is normal	
Format	CHECK	
Parameters	None	
Response	OK	
Example	Host: CHECK Slave: OK	Determine if communication is normal

2.2 Zero-bit mode

2.2.1 MODE_H

Item	Description	Remarks
Function	Sets the specified motor shaft to enter zero position mode and configures the corresponding zero position strategy	
Format	MODE_H obj method	
obj	Motor shaft sequence number, 0,1,2,3,4,5,6,7...	Starting from 0, maximum number of shafts and Model-dependent
method	0: Normal mode. Detects zero position upon hitting limit switch or zero position, then completes. 1: Zero position priority. If a limit switch is encountered first, reverse direction and search again until reaching a limit switch or zero position. If zero position is encountered first, complete immediately. 2: Closed-loop zero position. Determines the Z-axis reference number of the linear scale or encoder. . Closed-loop controllers use	
OK E1	OK E1 E2	
Example	Master: MODE_H 0 0 Slave: OK	

2.2.2 H_ACC_DEC

Item	Description	Remarks
Function	Set motion parameters for the specified motor shaft zero position	
Format	H_ACC_DEC obj acc dec	

obj	Motor shaft sequence number, 0,1,2,3,4,5,6,7…	Starting from 0, maximum number of shafts and Model-dependent
acc	Acceleration, unit step/s^2	
dec	Deceleration, unit step/s^2	
Response	OK E1 E2	
Example	Master: H_ACC_DEC 1 5000 8000 Slave: OK	Set the acceleration of the second axis to 5000, deceleration to 8000

2.2.3 H_V

Item	Description	Remarks
Function	Set the specified motor shaft to perform zero-position operation at the specified speed	Different zeroing strategies may result in may vary
Format	H_V obj v	
obj	Motor shaft sequence number, 0,1,2,3,4,5,6,7…	Starting from 0, maximum number of shafts and Model-dependent
v	Zero position speed, unit: steps/s	
Response	OK E1 E2	
Example	Master: H_V 0 -2000 Slave: OK	Set the first axis speed to 2000, reverse zero position

2.2.4 H_STOP

Item	Description	Remarks
Function	Decelerate and stop to specify motor shaft zero position action	Decelerate to stop; not an emergency stop
Format	H_STOP obj	
obj	Motor shaft sequence number, 0,1,2,3,4,5,6,7…	Starting from 0, maximum number of shafts and Model-dependent
Response	OK E1 E2	
Example	Master: H_STOP 0 Slave: OK	

2.3 Speed Mode (Constant Speed Mode)

2.3.1 MODE_V

Item	Description	Remarks
Function	Set the specified motor shaft to enter speed mode	
Format	MODE_V obj	
obj	Motor shaft number, 0,1,2,3,4,5,6,7…	Starting from 0, maximum number of shafts and Model-dependent
Response	OK E1	

	E2	
Example	Master: MODE_V 0 Slave: OK	

2.3.2 V_ACC_DEC

Item	Description	Remarks
Function	Set motion parameters for the specified motor shaft speed mode	
Format	V_ACC_DEC obj acc dec	
obj	Motor shaft sequence number, 0,1,2,3,4,5,6,7...	Starting from 0, maximum number of shafts and Model-dependent
acc	Acceleration, unit step/s^2	
dec	Deceleration, unit step/s^2	
Response	OK E1 E2	
Example	Master: V_ACC_DEC 1 5000 8000 Slave: OK	Set the acceleration of the second axis to 5000, deceleration to 8000

2.3.3 V_REL

Item	Description	Remarks
Function	Adjusts the speed of the specified motor shaft in speed mode relative to the current speed value	This is a relative value adjustment, applying a + or - offset to the current speed Yes, offset
Format	V_REL obj v	
obj	Motor shaft sequence number, 0,1,2,3,4,5,6,7...	Starting from 0, maximum number of shafts and Model-dependent
v	Speed, unit: steps/s	
Response	OK E1	

	E2	
Example	Master: V_REL 0 5000 Target: OK After 1S Main: V_REL 0 -3000 Target: OK	First accelerate to 5000, then decelerate 3000. If current speed 1000 , then first to 1000+5000=6000 target Accelerate at this speed After 1S, if accelerated to 5000, but has not reached 6000, decelerate by 3000, ultimately reaching 2000

2.3.4 V_ABS

term	Description	Remarks
Function	Adjusts the speed in the specified motor shaft speed mode. The value is the target speed. target speed	is an absolute value, and the required target Speed
Format	V_ABS obj v	
obj	Motor shaft sequence number, 0,1,2,3,4,5,6,7...	Starting from 0, maximum number of shafts and Model-dependent
v	Speed, unit: steps/s	
Response	OK E1 E2	
Example	Master: V_ABS 0 5000 Slave: OK After 1S Master: V_ABS 0 -3000 Slave: OK	First increase, target is positive rotation 5000, After 1S, target is reverse rotation 3000

2.3.5 V_STOP

Item	Description	Remarks
Function	Decelerate and stop to specify constant-speed operation of the motor shaft	Decelerate to stop; not an emergency stop
Format	V_STOP obj	
obj	Motor shaft sequence number, 0,1,2,3,4,5,6,7...	Starting from 0, maximum number of shafts and Model-dependent
Response	OK E1 E2	
Example	Master: V_STOP 0 Slave: OK	

2.4 Positioning Mode

2.4.1 MODE_P

Item	Description	Remarks

Function	Sets the specified motor shaft to enter positioning mode and configures the corresponding Positioning Strategy	
Format	MODE_P obj method	
obj	Motor shaft sequence number, 0,1,2,3,4,5,6,7...	Starting from 0, maximum number of shafts and Model-dependent
method	0: Open-loop mode, positioning based on transmitted pulses 1: Closed-loop mode, positioning based on received encoder or linear scale pulses	
Response	OK E1 E2	
Example	Master: MODE_P 0 0 Slave: OK	

2.4.2 P_ACC_DEC_V

Item	Description	Remarks
Function	Set motion parameters for the specified motor shaft position mode	
Format	P_ACC_DEC_V obj acc dec max_v	
obj	Motor shaft sequence number, 0,1,2,3,4,5,6,7...	Starting from 0, maximum number of shafts and Model-dependent
acc	Acceleration, unit $step/s^2$	
dec	Deceleration, unit $step/s^2$	
max_v	Maximum velocity, unit: step/s	Velocity at constant speed
Response	OK E1 E2	
Example	Master: P_ACC_DEC_V 2 5000 5000 20000 Slave: OK	Set the acceleration of the third axis to 5000, deceleration

		to 5000, maximum speed to 20000

2.4.3 P_FACTOR

Item	Description	Remarks
Function	Set output step count and ratio between grating scale/encoder pulses in closed-loop mode pulse ratio	
Format	P_FACTOR obj step pulse	
obj	Motor shaft number, 0,1,2,3,4,5,6,7…	Starting from 0, maximum number of shafts and Model-dependent
step	Output step count	
pulse	Grating scale/encoder pulse count	The controller automatically performs 4x

		. Note that the parameters marked on the linear scale or encoder are either quad-rated or non-quadruple frequency.
Response	OK E1 E2	
Example	Linear Stage Master: P_FACTOR 1 6400 4000 Slave: OK Rotary Stage Master: P_FACTOR 1 64000 4000 Slave: OK	For example, a stepper system with 32 microsteps, Linear stage with 1mm pitch, encoder resolution 1μm per division, then 1mm corresponds to 6400 output pulses, 1mm corresponds to 1000 grating scale outputs, and after quadruple frequency conversion, 4000 32 microsteps, 10:1 gear ratio rotary stage with encoder mounted on the , one full rotation of the stage is 6400*10=64000, encoder For a 1000-line device, the quadruple frequency results in 4000 pulses.

2.4.4 P_REL

term	Description	Remarks

Function	Adjust the target position in the specified motor shaft position mode relative to the current position value	This is a relative value adjustment, based on the current position with a ± Yes, offset
Format	P_REL obj p	
obj	Motor shaft sequence number, 0,1,2,3,4,5,6,7…	Starting from 0, maximum number of shafts and Model-dependent
p	Position, open-loop unit step, change unit is pulse	
Response	OK E1 E2	
Example	Master: P_REL 0 5000 Slave: OK After 1S Master: P_REL 0 -3000 Slave: OK	First, set the current position +5000 as the motion target. After 1 second, set the current position minus 3000 as the target. If the current position is 1000, it will first move toward the target position 1000+5000=6000 target position. After 1 second, if it has moved to 5000 but not yet reached 6000,

		subtract 3000 again, setting the final target position is 2000.

2.4.5 P_ABS

Item	Description	Remarks
Function	Adjust the target position in positioning mode for the specified motor shaft	
Format	P_ABS obj p	
obj	Motor shaft sequence number, 0,1,2,3,4,5,6,7…	Starting from 0, maximum number of shafts and Model-dependent
v	Position, open-loop unit: step, increment unit: pulse	
Response	OK E1 E2	
Example	Master: P_ABS 0 5000 Slave: OK After 1S Master: P_ABS 0 -3000 Slave: OK	First acceleration, target position is 5000, After 1 second, the target position is 3000. The controller automatically adjusts direction and speed.

2.4.6 P_STOP

Item	Description	Remarks
Function	Decelerate and stop the specified motor shaft positioning action	Decelerate and stop; not an emergency stop
Format	P_STOP obj	
obj	Motor shaft sequence number, 0,1,2,3,4,5,6,7…	Starting from 0, maximum number of shafts and Model-dependent

Response	OK E1 E2	
Example	Master: P_STOP 0 Slave: OK	

2.5 Multi-axis Interpolation

2.5.1 L_ACC_DEC_V

Item	Description	Remarks

Function	Set motion parameters for specified multi-axis interpolation modules	Spatial Linear Interpolation
Format	L_ACC_DEC_V obj acc dec max_v	
obj	Interpolation module number, 0,1,2,3…	Starting from 0, maximum number depends on model; multiple interpolation modules may be present modules may operate simultaneously
acc	Acceleration, unit: $step/s^2$	
dec	Deceleration, unit $step/s^2$	
max_v	Maximum velocity, unit: step/s	Maximum shaft speed at constant velocity
Response	OK E1 E2	
Example	Master: L_ACC_DEC_V 0 5000 5000 10000 Responder: OK	Set the acceleration of the first interpolation module to 5000 and the deceleration to 5000, and maximum velocity to 10000

2.5.2 L_REL

Item	Description	Remarks
Function	Performs multi-axis interpolation actions for the specified module, where the target value is relative to the current position.	This is a relative value adjustment. The target value in the current based on the current position, with adjustments of ± offset. During interpolation, parameters cannot be modify parameters during interpolation; they can only be modified after

		stopping modify them.
Format	P_REL obj m0 m1 m2 t0 t1 t2	The number of m's and the number of t's must equal.
obj	Interpolation module serial number, 0,1,2,3,…	Starting from 0, maximum number depends on Model-dependent; multiple interpolation modules can operate simultaneously
m	Motor shaft sequence number, 0,1,2,3,4,5,6,7…	Starting from 0, maximum number of axes depends on model Model-dependent, supports up to 8 axes for interpolation
t	Position adjustment amount, unit: step Adjustment relative to the current position	Only supports open-loop interpolation
Response	OK E1 E2	
Example	Master: 0 3 1 5000 -2000 L_REL E2: OK Master: 1 0 2 4 -3000 2000 8000 L_REL Slave: OK	Axes 3 and 1 utilize interpolation mode Block 0 for interpolation motion, each offset relative to the current position Axes 0, 2, and 4 utilize Interpolation Module 1 for interpolation operations

		relative to their respective current bit offset

2.5.3 L_ABS

Item	Description	Remarks
Function	Performs multi-axis interpolation actions using the specified interpolation module, where the target value represents the final destination.	Performs absolute value adjustment, automatically calculating offsets based on the current position. **Parameters cannot be modified during interpolation; they can only be adjusted after stopping.** .
Format	P_ABS obj m0 m1 m2 t0 t1 t2	The number of m's and the number of t's must be equal. equal
obj	Interpolation module number, 0,1,2,3,…	Starting from 0, maximum number depends on model; multiple interpolation modules operating simultaneously
m	Motor shaft sequence number, 0,1,2,3,4,5,6,7…	Starting from 0, maximum number of axes depends on model-dependent, supporting up to 8 axes for interpolation
t	Position adjustment amount, unit: step Adjustment relative to the current position	Only supports open-loop interpolation
Response	OK E1	

	E2	
Example	Subject: L_ABS 0 3 1 5000 -2000 Subject: OK 1 0 2 4 -3000 2000 8000 Subject: L_ABS Re: OK	Axes 3 and 1 utilize interpolation mode Block 0 for interpolation motion, aiming to move to (5000, -2000). Axes 0, 2, and 4 utilize interpolation module 1 to move to the target coordinates (-3000, 2000, 8000).

2.5.4 L_STOP

item	Description	Remarks
Function	Decelerate and stop specified multi-axis interpolation motion	Decelerate and stop; not an emergency stop
Format	L_STOP obj	
obj	Interpolation module number, 0,1,2,3,...	Starting from 0, maximum number related to model

		Model-dependent; multiple interpolation modules operating simultaneously
Response	OK E1 E2	
Example	Master: L_STOP 0 Slave: OK	

2.6 Arc Interpolation

2.6.1 C_ACC_DEC_V

Item	Description	Remarks
Function	Set motion parameters for the specified arc interpolation module	
Format	C_ACC_DEC_V obj acc dec max_v	
obj	Interpolation module number, 0,1,2,3…	Starting from 0, maximum number depends on model; multiple interpolation modules may be present modules may operate simultaneously
acc	Acceleration, unit: step/s^2	
dec	Deceleration, unit step/s^2	
max_v	Maximum speed, unit: steps/s	Linear velocity at constant speed
Response	OK E1 E2	
Example	Master: C_ACC_DEC_V 0 5000 5000 10000 Slave: OK	Set the acceleration of the first interpolation module to 5000 and the deceleration to

		5000, and maximum velocity to 20000

2.6.2 C_CW_REL

Item	Description	Remarks
Function	Performs arc interpolation using the specified arc interpolation module, where the target value is adjusted relative to the current position of all axes involved in the interpolation. Clockwise motion	Relative value adjustment, based on the current position with a ± offset Offset value Parameters cannot be modified during interpolation; they can only be adjusted after stopping. Modify
Format	C_CW_REL obj m0 m1 t0 t1 r	r is the absolute value, not the relative value
obj	Interpolation module number, 0, 1, 2, 3, …	Starting from 0, maximum number and

		Model-dependent; multiple interpolation modules may operate simultaneously
m	Motor shaft serial number, 0,1,2,3,4,5,6,7...	Starting from 0, maximum number of shafts Model-dependent
t	Position adjustment amount, unit: step Adjustment relative to the current position	Supports open-loop interpolation only
r	Arc radius	**When r is positive, it is a minor arc When r is negative, it is a superior arc**
Response	OK E1 E2	
Example	Main: 1 5000 -2000 C_CW_REL 10000 0 3 Recipient: OK Main: 0 2 -3000 2000 - C_CW_REL 1 10000 Target: OK	Axes 3 and 1 utilize interpolation mode Block 0 for interpolation motion, each relative to the current position Move, arc radius is 10000, traverse the inferior arc Axis 0 and Axis 2 using interpolation mode Block 1 performs interpolation motion, moving relative to its current position with a displacement of with an arc radius of 10000. Optimal Arc

2.6.3 C_CCW_REL

Item	Description	Remarks
Function	Specifies the arc interpolation module to perform arc interpolation operations, where the target value is adjusted relative to the current positions of all axes involved in the interpolation. Counterclockwise motion	Relative value adjustment, based on the current position with a ± Offset value Parameters cannot be modified during interpolation; they can only be adjusted after stopping. Modify
Format	C_CW_REL obj m0 m1 t0 t1 r	r is the absolute value, not the relative value
obj	Interpolation module number, 0, 1, 2, 3, ...	Starting from 0, maximum number depends on model; multiple interpolation modules may be present Modules can operate simultaneously
m	Motor shaft sequence number, 0,1,2,3,4,5,6,7...	Starting from 0, maximum number of shafts depends on model
t	Position adjustment amount, unit: step Adjustment relative to the current position	Supports open-loop interpolation only
r	Arc radius	When r is positive, it is a minor arc When r is negative, it is a superior arc

Response	OK E1 E2	
Example	Master: 0 1 5000 -2000 C_CW_REL 10000 3 Recipient: OK Main: 1 2 -3000 2000 - C_CCW_REL 0 10000 Target: OK	Axes 3 and 1 utilize interpolation mode Block 0 for interpolation motion, moving relative to their current positions by Move, arc radius is 10000, traverse the inferior arc Axes 0 and 2 perform interpolation motion using interpolation mode Block 1 for interpolation movement, each moving relative to the current position by with an arc radius of 10000, Optimal Arc

2.6.4 C_CW_ABS

item	Description	Remarks
Function	The specified circular interpolation module performs circular interpolation operations, where the target value is an absolute target value. Clockwise motion	The target is adjusted to an absolute value. **Parameters cannot be modified during interpolation; they can only be adjusted after stopping.** **Modify**
Format	C_CW_REL obj m0 m1 t0 t1 r	r is the absolute value, not the

		relative value
obj	Interpolation module number, 0, 1, 2, 3, …	Starting from 0, maximum number depends on model; multiple interpolation modules may be present Modules can operate simultaneously
m	Motor shaft sequence number, 0,1,2,3,4,5,6,7…	Starting from 0, maximum number of shafts depends on model-dependent
t	Position adjustment amount, unit: step Adjustment relative to the current position	Supports open-loop interpolation only
r	Arc radius	**When r is positive, it is a minor arc** **When r is negative, it is a superior arc**
Response	OK E1 E2	
Example	Master: C_CW_ABS 10000 0 3 1 5000 -2000 Recipient: OK Main: C_CW_ASB 0 2 -3000 2000 -1 10000 Target: OK	Axes 3 and 1 utilize interpolation mode Block 0 for interpolation movement, targeting (5000, -2000) with a circular with a radius of 10000, following the inferior arc Axes 0 and 2 perform interpolation motion using interpolation mode Block 1 for interpolation motion,

		target coordinates (-3000, 2000) arc radius of 10000, using optimal arc

2.6.5 C_CCW_ABS

Option	Description	Remarks
Function	Performs arc interpolation using the specified arc interpolation module, where the target value is an absolute target value Counterclockwise motion	Target is adjusted to absolute value **Parameters cannot be modified during interpolation; they can only be adjusted after stopping. Modify**
Format	C_CW_REL obj m1 t0 t1 r m0	r is the absolute value, not the relative value
obj	Interpolation module number, 0, 1, 2, 3, …	Starting from 0, maximum number depends on model; multiple interpolation modules may be present Modules can operate simultaneously
m	Motor shaft sequence number, 0,1,2,3,4,5,6,7…	Starting from 0, maximum number of shafts depends on model-dependent
t	Position adjustment amount, unit: step Adjustment relative to the current position	Supports only open-loop interpolation
r	Arc radius	**When r is positive, it is a minor arc** **When r is negative, it is a superior arc**
Response	OK E1	

	E2	
Example	Master: C_CCW_ABS 0 3 1 5000 -2000 10000 Subject: OK Master: C_CCW_ASB 1 0 2 -3000 2000 - 10000 Rejected: OK	Axes 3 and 1 utilize interpolation mode Block 0 for interpolation movement, target position (5000, -2000) arc radius 10000, using the inferior arc Axes 0 and 2 use interpolation mode Block 1 for interpolation movement, targeting (-3000, 2000)with an arc radius of 10000, following the inferior arc

2.6.6 C_STOP

Item	Description	Remarks
Function	Decelerate and stop the specified circular interpolation motion	Decelerate to stop; not an emergency stop
Format	C_STOP obj	
obj	Interpolation module number, 0,1,2,3,…	Starting from 0, maximum number related to model Model-dependent, multiple interpolation

		modules operating simultaneously
Response	OK E1 E2	
Example	Master: C_STOP 0 Slave: OK	

2.7 Emergency Stop

2.7.1 HALT_ONE

Item	Description	Remarks
Function	Emergency stop specified motor shaft operation	Emergency stop pulse output without deceleration may cause mechanical vibration
Format	HALT_ONE obj	
obj	Motor shaft sequence number, 0,1,2,3,4,5,6,7...	Starting from 0, maximum number of shafts and Model-dependent
Response	OK E1 E2	
Example	Master: HALT_ONE 0 Slave: OK	

2.7.2 HALT_ALL

Item	Description	Remarks
Function	Emergency stop all motor shaft movements	Emergency stop pulse output without deceleration may cause mechanical vibration
Format	HALT_ALL	

Response	OK E1 E2	
Example	Master: HALT_ONE 0 Slave: OK	

2.7.3 HALT_L

Item	Description	Remarks
Function	Emergency stop of specified multi-axis interpolation motion	Emergency stop pulse output, without deceleration, which may cause mechanical vibration
Format	HALT_L obj	
obj	Interpolation module number, 0,1,2,3,...	Starting from 0, maximum number related to model Model-dependent; multiple interpolation modules can operate simultaneously
Response	OK E1 E2	
Example	Master: HALT_L 0 Slave: OK	

2.7.4 HALT_C

Item	Description	Remarks
Function	Emergency stop of specified circular interpolation motion	Emergency stop pulse output, deceleration may cause mechanical vibration.
Format	HALT_C obj	
obj	Interpolation module number, 0,1,2,3,...	Starting from 0, maximum number depends on model; multiple interpolation modules may be present modules may operate simultaneously

Response	OK E1 E2	
Example	Master: HALT_C 0 Slave: OK	

2.8 Coordinate Adjustment

2.8.1 SET_P

Item	Description	Remarks
Function	Adjusts the coordinate position of the specified motor shaft, equivalent to the coordinate system Translation	
Format	SET_P obj p	
obj	Motor shaft sequence number, 0,1,2,3,4,5,6,7...	Starting from 0, maximum number of shafts and

		Model-dependent
p	New coordinates, unit: step	
Response	OK E1 E2	
Example	Master: SET_P 0 0 Target: OK	After power-on zeroing, the position of Axis 0 is set to coordinate origin 0. Then, move forward by 1000 units step. The current position is 1000. This command then sets the current position to 0, establishing it as the new coordinate origin. All subsequent target positions will be calculated relative to this point.

2.8.2 SET_ENCODER

Item	Description	Remarks
Function	Adjusts the closed-loop coordinate position of the specified motor shaft, equivalent to a coordinate system translation. Coordinate System Translation	
Format	SET_ENCODER obj p	
obj	Motor shaft sequence number, 0,1,2,3,4,5,6,7…	Starting from 0, maximum number of shafts and Model-dependent
p	New coordinates, unit: pulse	
Response	OK E1 E2	

Example	Master: SET_ENCODER 0 1000 Slave: OK	Unconditionally adjust the current encoder grating scale count to 1000 **This command is not suitable for absolute encoders.**

2.9 Motor Shaft Status

2.9.1 GET_RUN

Item	Description	Remarks
Function	Query the operational status of the specified motor shaft	

Format	GET_RUN obj	
obj	Motor shaft serial number, 0,1,2,3,4,5,6,7...	Starting from 0, maximum number of shafts and Model-dependent
Response	0: Stopped 1: Running E1 E2	
Example	Master: GET_RUN 0 Slave: 1	

2.9.2 GET_NEG

Item	Description	Remarks
Function	Queries the negative limit status of the specified motor shaft	
Format	GET_NEG obj	
obj	Motor shaft serial number, 0,1,2,3,4,5,6,7...	Starting from 0, maximum number of shafts and Model-dependent
Response	0: Limit switch inactive 1: Limit effective E1 E2	
Example	Master: GET_NEG 0 Slave: 1	

2.9.3 GET_POS

Item	Description	Remarks
Function	Query the positive limit status of the specified motor shaft	
Format	GET_POS obj	
obj	Motor shaft serial number: 0, 1, 2, 3, 4, 5, 6, 7...	Starting from 0, maximum number of shafts and Model-dependent

Response	0: Limit switch inactive 1: Limit effective E1 E2	
Example	Master: GET_POS 0 Slave: 1	

2.9.4 GET_ZERO

Item	Description	Remarks
Function	Query the zero position status of the specified motor shaft	

Format	GET_ZERO obj	
obj	Motor shaft sequence number, 0,1,2,3,4,5,6,7...	Starting from 0, maximum number of shafts and Model-dependent
Response	0: Zero position invalid 1: Zero position valid E1 E2	
Example	Master: GET_ZERO 0 Slave: 1	

2.9.5 GET_P

Item	Description	Remarks
Function	Query the coordinate position of the specified motor shaft	
Format	GET_P obj	
obj	Motor shaft serial number, 0,1,2,3,4,5,6,7...	Starting from 0, maximum number of shafts and Model-dependent
Response	nnnnnn Number For example 34587 E1 E2	
Example	Main: GET_P 0 Target: 34587	

2.9.6 GET_V

Item	Description	Remarks
Function	Query the current speed of the specified motor shaft	
Format	GET_V obj	
obj	Motor shaft serial number: 0, 1, 2, 3, 4, 5, 6, 7...	Starting from 0, maximum number of shafts and Model-dependent
Response	nnnnnn Number For example 34587 E1 E2	

Example	Main: GET_V 0 Target: 20000	

2.9.7 GET_ENCODER

Item	Description	Remarks
Function	Query the encoder/ruler count of the specified motor shaft	
Format	GET_ENCODER obj	

obj	Motor shaft serial number, 0,1,2,3,4,5,6,7...	Starting from 0, maximum number of shafts and Model-dependent
Response	nnnnnn Number For example 34587 E1 E2	
Example	Main: GET_ENCODER 0 Target: 34587	

2.9.8 GET_L_RUN

Item	Description	Remarks
Function	Query the operational status of the specified multi-axis interpolation module	
Format	GET_L_RUN obj	
obj	Interpolation module serial number, 0,1,2,3,...	Starting from 0, maximum number depends on model; multiple interpolation modules may be present modules may operate simultaneously
Response	0: Stopped 1: Running E1 E2	
Example	Master: GET_L_RUN 0 Slave: 1	

2.9.9 GET_C_RUN

Item	Description	Remarks
Function	Query the operational status of the specified circular arc interpolation module	
Format	GET_C_RUN obj	

obj	Interpolation module serial number, 0,1,2,3,…	Starting from 0, maximum number depends on model; multiple interpolation modules may be present modules operating simultaneously
Response	0: Stopped 1: Running E1 E2	
Example	Master: GET_C_RUN 0 Slave: 1	

2.9.10 GET_MODE

Item	Description	Remarks
Function	Query the operating mode of the specified motor shaft	
Format	GET_MODE obj	
obj	Motor shaft serial number, 0,1,2,3,4,5,6,7…	Starting from 0, maximum number of shafts and Model-dependent
Response	0: Zero Position Mode 1: Constant Speed Mode 2: Positioning mode 3: Multi-axis Interpolation 4: Arc Interpolation E1 E2	Default setting after power-on is 2 If axes that have used multi-axis interpolation or circular interpolation are not performing interpolation, they must be restored to the required operating mode via command.
Example	Subject: 0 GET_MODE Target: 2	

2.10 General-Purpose Input/Output

2.10.1 SET_OUT

Item	Description	Remarks
Function	Adjust the output state of the specified optically isolated output	
Format	SET_OUT obj v	
obj	Optically isolated output sequence number, 0,1,2,3,4,5,6,7…	Starting from 0, maximum number and type Related to serial number

v	New state, high/low level 0: Off 1: Open optocoupler	Optocoupler output depends on actual wiring configuration
Response	OK E1 E2	
Example	Master: 0 0 SET_OUT Responder: OK	

2.10.2 SET_OC

Item	Description	Remarks
Function	Adjust the output state of the specified OC output	
Format	SET_OC obj v	
obj	OC Output sequence number, 0,1,2,3,4,5,6,7...	Starting from 0, maximum number and type Related to serial number

v	New state, high/low level 0: Transistor off 1: Turn transistor on	Output depends on actual wiring configuration
Response	OK E1 E2	
Example	Master: SET_OC 0 0 Slave: OK	

2.10.3 GET_IN

Item	Description	Remarks
Function	Query the status of a specified optically isolated input	Limit and zero positions are shared with certain inputs; these functions can be disabled via the configuration tool Function
Format	GET_IN obj	
obj	Optically isolated input sequence number, 0,1,2,3,4,5,6,7...	Starting from 0, maximum number and type Related to response number
Response	0: External low level 1: External High Level E1 E2	is related to the externally connected circuit; when not connected, it defaults to high level
Example	Main: GET_IN 0 Slave: 1	

2.10.4 GET_OUT

Item	Description	Remarks
Function	Query the current status of the specified optically isolated output	
Format	GET_OUT obj	
obj	Optically isolated output serial number: 0, 1, 2, 3, 4, 5, 6, 7...	Starting from 0, maximum number and type

		Related to serial number
Response	0: Optocoupler off 1: Optocoupler ON E1 E2	
Example	Master: GET_OUT 0 Slave: 1	

2.10.5 GET_OC

Item	Description	Remarks

Function	Query the current status of the specified OC	
Format	GET_OC obj	
obj	OC output sequence number: 0, 1, 2, 3, 4, 5, 6, 7...	Starts at 0, maximum number and type Related to the number
Response	0: Transistor off 1: Transistor open E1 E2	
Example	Master: GET_OC 0 Sub: 1	